

Business Requirements Document (BRD)

Multi-Tenant SaaS API — business problem, objectives, stakeholders, requirements, constraints, risks and future expansion

1. Business Overview

The proposed product is an organization-centric SaaS platform for project and task management. The initial business purpose is learning and portfolio development, while the architecture is intentionally designed so the product can become commercial later if desired.

2. Business Problem

Organizations need a centralized way to manage projects, tasks and collaboration. The product aims to provide organization-level isolation, role-based access and operationally reliable APIs as the foundation for this capability.

3. Business Opportunity

The initial opportunity is not immediate revenue generation. It is to create a realistic SaaS foundation that demonstrates the capabilities expected of a production backend engineer and can later be extended into a commercial offering if product-market validation warrants it.

4. Business Objectives

- Deliver a coherent multi-tenant project/task management product.
- Demonstrate secure and scalable backend engineering.
- Build reusable foundations for future commercial features.
- Avoid premature investment in features that are only necessary after commercialization.
- Create a product that can be incrementally expanded rather than rewritten.

5. Stakeholders

Stakeholder	Interest
Product owner / learner	Learning value, engineering depth, maintainability and portfolio quality.
Organization owner	Control over organization membership, projects and permissions.
Organization admin	Member administration and organization-level operations.
Project manager	Project execution, task management and collaboration.
Members	Efficient task execution and collaboration.
Viewers	Controlled read access.
Future commercial stakeholders	Potential billing, compliance, identity, storage and enterprise capabilities.

6. Business Requirements

ID	Business requirement	Description
BR-001	Organization isolation	Each organization's data and operations must be logically isolated.
BR-002	Role-based administration	Organizations must be able to control what users can do through roles and permissions.
BR-003	Project execution	Organizations must be able to create projects and manage project work.
BR-004	Task collaboration	Teams must be able to create, assign and track tasks.
BR-005	Communication	Users must be able to collaborate through comments and notifications.
BR-006	Operational reliability	The platform must provide sufficient observability and resilience to diagnose and manage failures.
BR-007	Scalability path	The design should support growth toward approximately 100K users without a fundamental rewrite.
BR-008	Future extensibility	Commercial features can be introduced later without invalidating the core domain model.

7. Business Success Criteria

- A complete organization-to-project-to-task workflow is usable end to end.
- No known cross-tenant access path exists in tested core workflows.
- Role permissions are understandable and consistently enforced.
- The API can support normal frontend application workflows without requiring direct database access.
- Operational failures can be investigated through logs, metrics and traces.
- The system has automated validation and regression coverage.
- The architecture can accept future commercial capabilities such as file storage, 2FA and billing.

8. Business Constraints

- This is initially a learning/portfolio project, so engineering complexity must remain proportional to the learning objective.
- Commercial features are not allowed to dominate the initial roadmap.
- File upload/storage is explicitly deferred.
- The initial membership model is organization-based.
- The primary API consumer is a web frontend.

9. Business Dependencies

- PostgreSQL for durable application data.
- Redis for rate limiting, caching, session/token management and distributed locks.
- Kafka for preferred asynchronous/event-driven processing.
- Email provider for notification delivery.
- Prometheus/Grafana/OpenTelemetry for observability.
- Container/runtime and CI/CD platform to be selected later.

10. Business Risks

Risk	Business impact	Response
Scope expansion	The product contains many infrastructure capabilities that can distract from the core user journey.	Use staged priorities and phase gates.
Commercial premature optimization	Building billing/SSO/storage too early can slow learning.	Keep them in P3.
Security risk	Multi-tenancy makes authorization errors potentially severe.	Treat tenant isolation as a P0 business requirement.
Technology-driven design	Adding Kafka/Redis without a real use case can create unnecessary complexity.	Introduce each technology through a concrete product/operational requirement.

11. Future Commercial Expansion

If the product later becomes a serious commercial SaaS, the business can expand into billing/subscriptions, usage limits, actual file storage, 2FA, OAuth, SSO, audit logs, advanced analytics, richer membership models and enterprise capabilities. These are deliberately outside the current business baseline.